



Feedback form for the course

Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

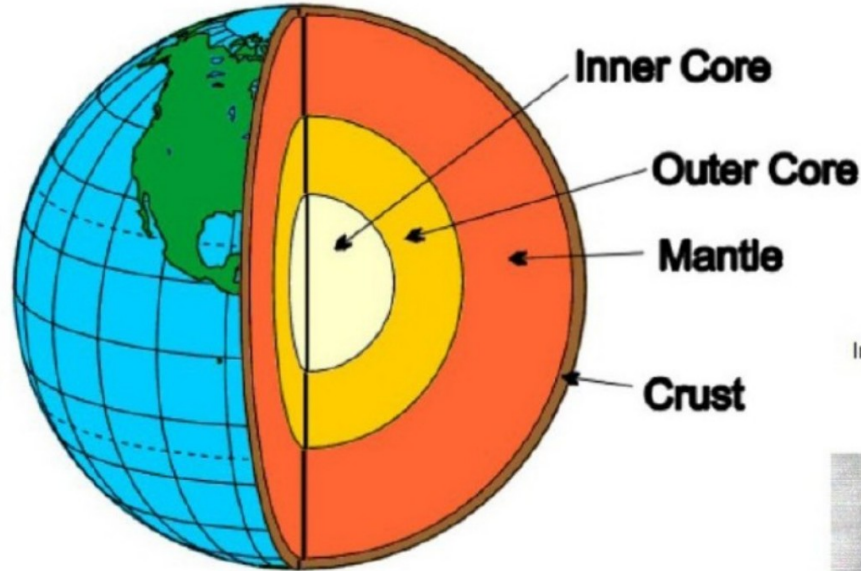
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Delay time tomography

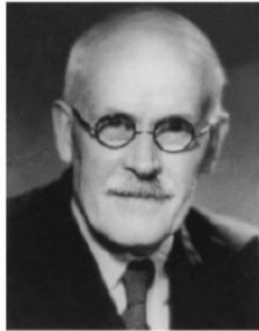
From last lecture



Richard Oldham 1858 – 1936



Inge Lehmann 1888 – 1993



Harold Jeffreys 1891 – 1989

Of all regions of the earth none invites speculation more than that which lies beneath our feet, and in none is speculation more dangerous; yet, apart from speculation, it is little that we can say regarding the constitution of the interior of the earth ... Many theories of the earth have been propounded at different times: the central substance of the earth has been supposed to be fiery, fluid, solid, and gaseous in turn, till geologists have turned in despair from the subject, and become inclined to confine their attention to the outermost crust of the earth, leaving its centre as a playground for mathematicians.

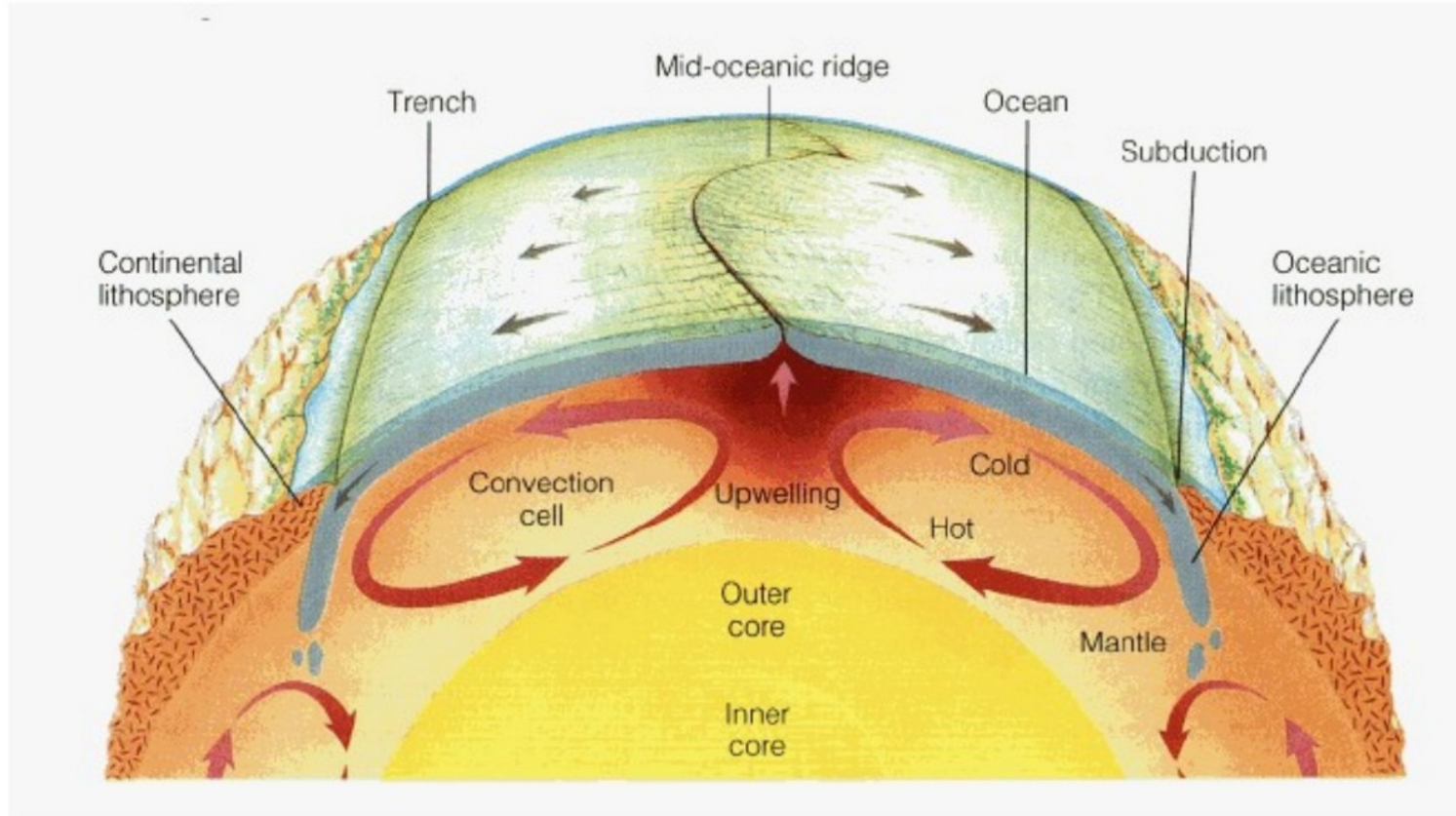
Richard Oldham 1906



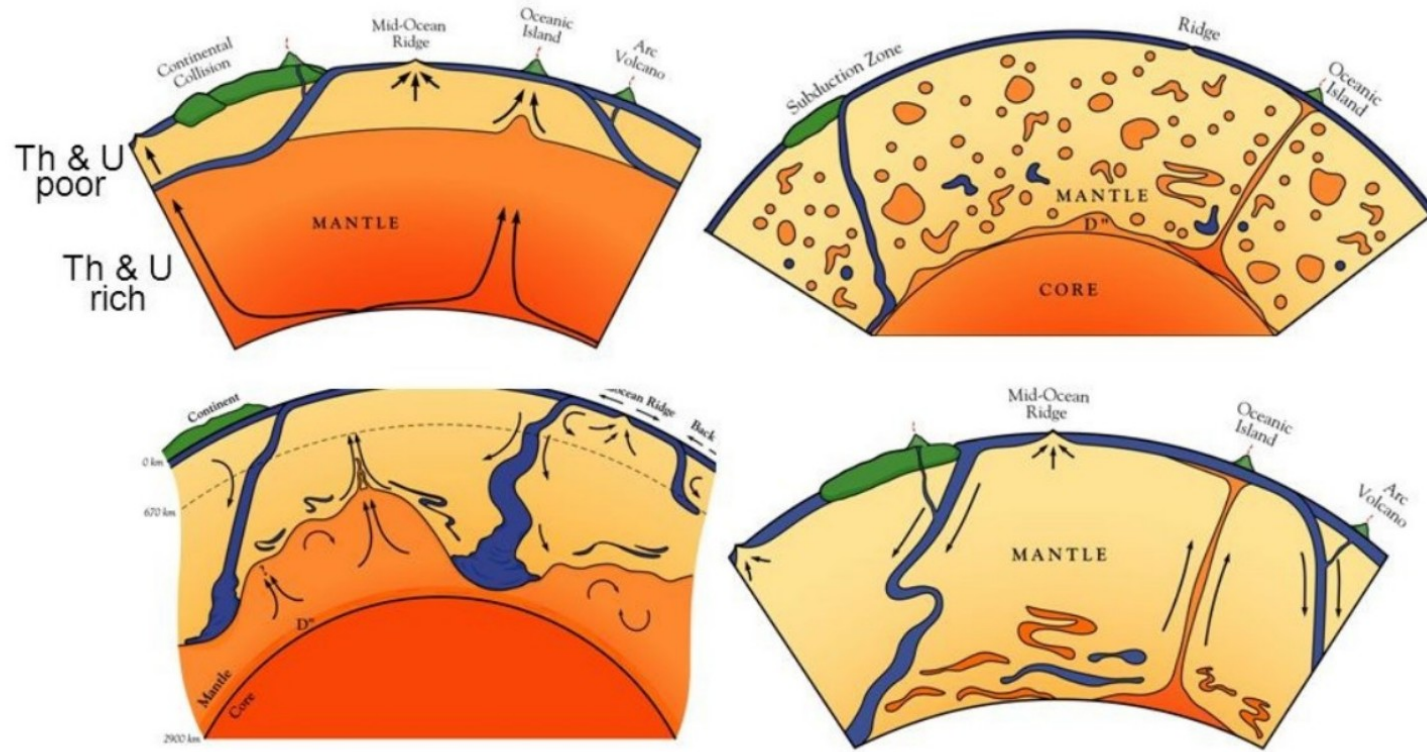
Beno Gutenberg 1889 – 1960

Lateral velocity variations

Mantle convection

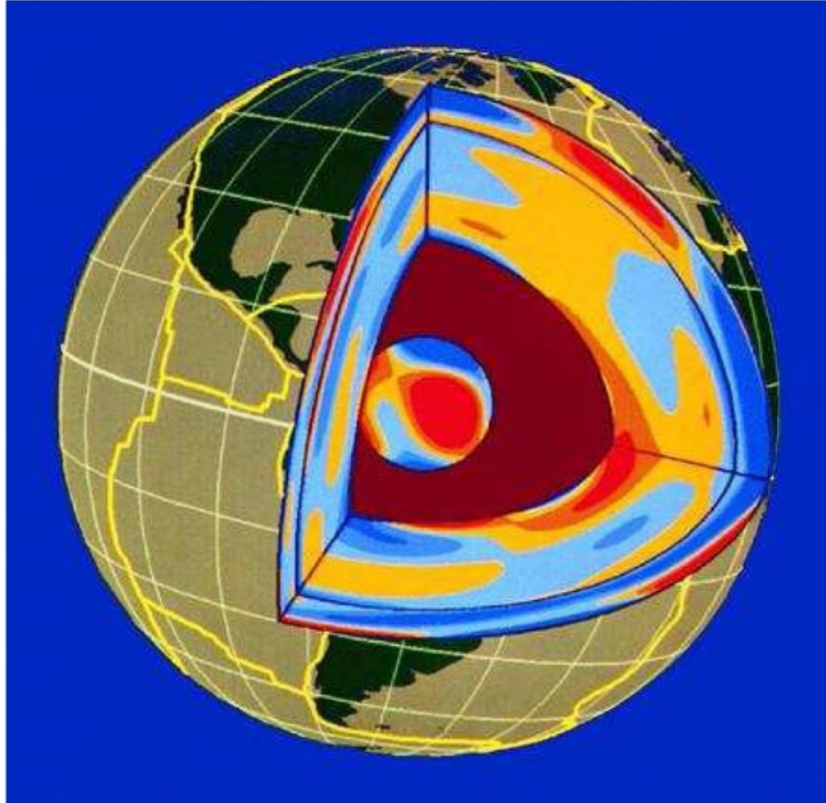


Mantle convection



Can we do better than drawing cartoons?

Seismic tomography



Dziewonski & Woodhouse (1989)

Quantitative models for lateral variations within the Earth derived from seismological observations

Most models show perturbations in elastic wave speeds relative to a spherically symmetric reference model

Other parameters, (e.g., density, attenuation can be sought with more difficulty)

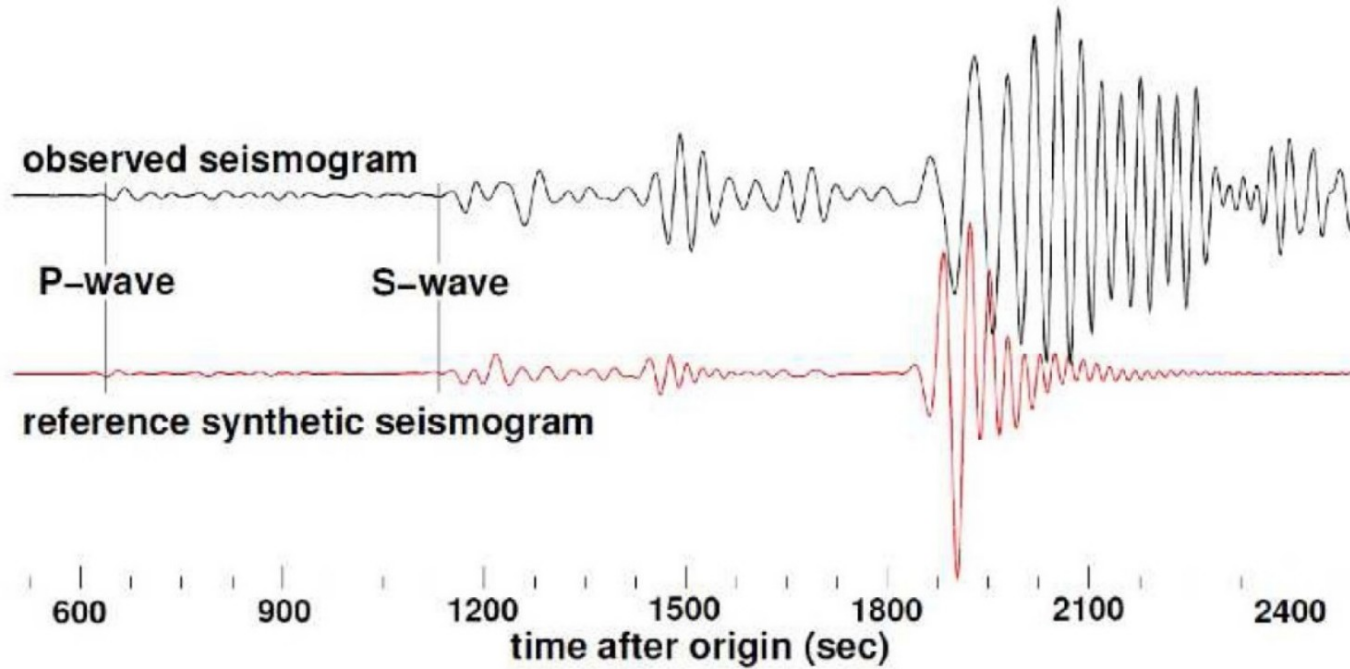
Simple interpretation:

Red = slow = hot

Blue = fast = cold

Setting up the inverse problem

Delay time measurements



$$\delta T = T - T_0.$$

Ray theoretic delay times

Rays determined through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

Once the ray path is known, the travel time can be written:

$$T = \int_{\text{ray}} \frac{1}{v} ds,$$

If velocity variations are sufficiently small, we can seek to linearise:

$$T(v + \delta v) = T(v) + \langle DT(v) | \delta v \rangle + O(\delta v^2).$$

Ray theoretic delay times

Using Fermat's principle, we find that:

$$\delta T = - \int_{\text{ray}} \frac{\delta v}{v^2} ds,$$

First-order contribution from varying ray geometry vanishes, and so we need only integrate the velocity perturbation along the reference ray path

Functional derivative identified by using the standard inner product:

$$\langle u | w \rangle = \int_M u w d^3 \mathbf{x},$$

Note that the functional derivative is singular on the ray path and zero elsewhere

Finite-dimensional parametrisation

For the i th delay time measurement:

$$\delta T_i = - \int_{\text{ray}_i} \frac{\delta \alpha}{\alpha^2} \mathrm{d}s + e_i.$$

Expand the velocity in a finite-dimensional basis:

$$\delta \alpha(\mathbf{x}) = \sum_{j=1}^m \delta \alpha_j \varphi_j(\mathbf{x}).$$

Choice of basis functions determined by assumptions about the Earth and the potential resolution of the data as set by its dominant frequencies

Reduction to linear algebra

Using the model parametrisation, we now have:

$$\delta T_i = \sum_{j=1}^m \left(- \int_{\text{ray}_i} \frac{\varphi_j}{\alpha^2} \mathrm{d}s \right) \delta \alpha_j + e_i,$$

We can now use a familiar matrix-vector notation:

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e},$$

where we have set:

$$A_{ij} = - \int_{\text{ray}_i} \frac{\varphi_j}{\alpha^2} \mathrm{d}s,$$

Solving the inverse problem

The null space

Tomographic problem with perfect data:

$$\mathbf{d} = \mathbf{A}\mathbf{m}.$$

Is there unique model that solves this problem?

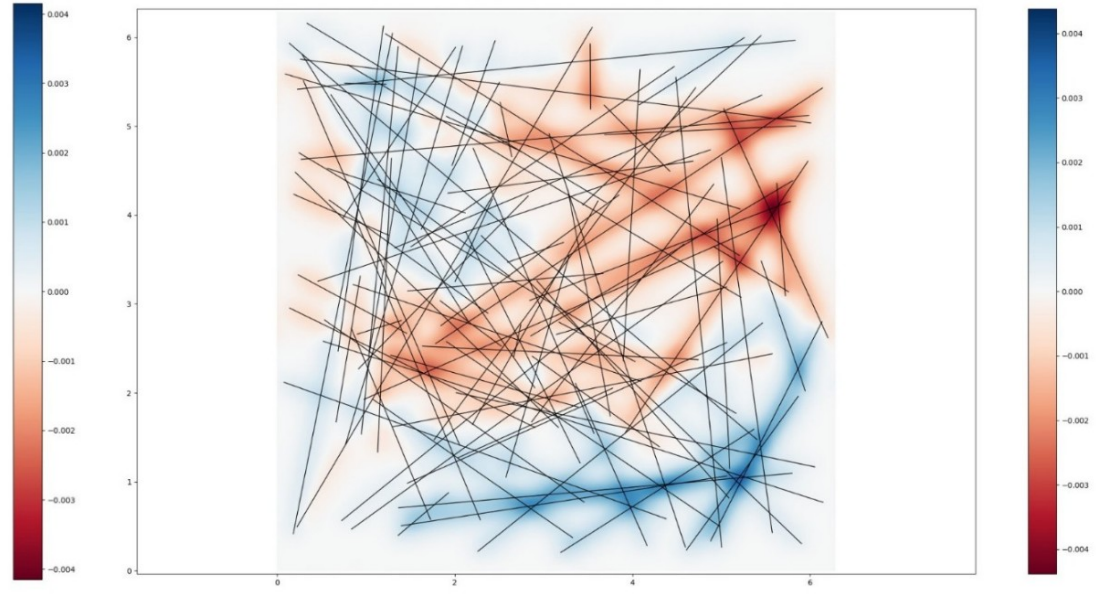
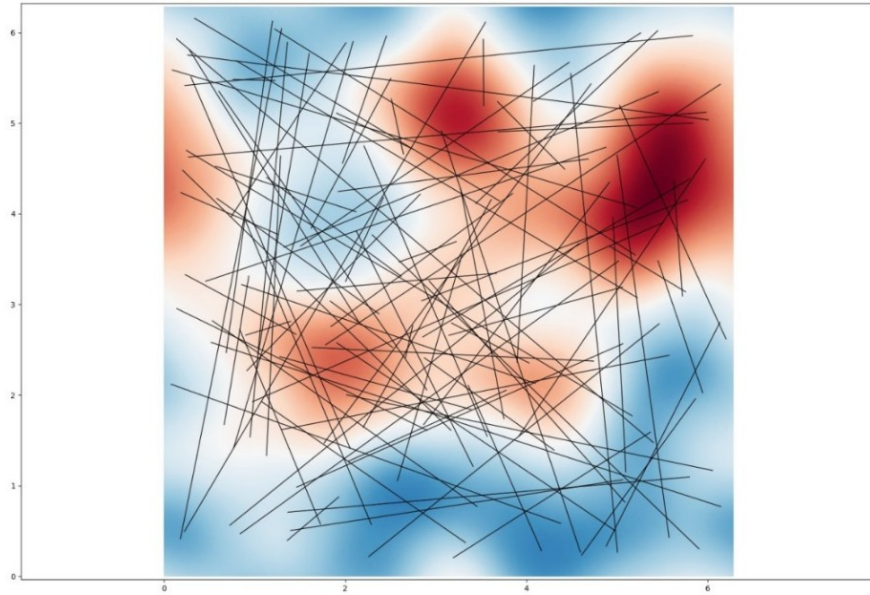
The **null space** for the matrix **A** comprises those vectors satisfying:

$$\mathbf{A}\mathbf{m}_0 = \mathbf{0}$$

This is a linear subspace. If non-trivial (not just the zero-vector), then the linear system cannot have unique solutions

The null space is always non-trivial for under-determined problems $M > N$

The null space



Lack of ray coverage is the most obvious ray null spaces are generated

Least squares solutions

An obvious way to proceed is through a **least-squares solution**

We define the least-squares misfit through:

$$J(\mathbf{m}) = \frac{1}{2} \sum_{i=1}^n \frac{[(\mathbf{A}\mathbf{m})_i - d_i]^2}{\sigma_i^2}$$

In matrix notation, this becomes:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}).$$

If there is a non-trivial null-space there is **no unique minimum!**

Regularisation

The idea is to penalise models with undesirable properties within the inversion

Suppose we prefer models with a small norm:

$$\int_M \delta\alpha^2 d^3\mathbf{x}.$$

Relative to our model parametrisation, this becomes:

$$\int_M \delta\alpha^2 d^3\mathbf{x} = \mathbf{m}^T \mathbf{B} \mathbf{m},$$

$$B_{jk} = \int_M \varphi_j \varphi_k d^3\mathbf{x}.$$

Regularised least squares

We modify the least squares misfit to incorporate regularisation:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}\mathbf{m}^T \mathbf{B}\mathbf{m},$$

Models with a small value of J now represent a balance between fitting the data well and keeping the chosen model property small

This balance is reflected in the choice of the **trade-off parameter**

Many choices of regularisation possible such as **norm or gradient damping**

Solving the optimisation problem

Misfit is quadratic, and so has at most one minimum value when:

$$\nabla J(\mathbf{m}) = \mathbf{0}$$

Through a calculation (detailed in the handout) this implies:

$$(\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B}) \mathbf{m} = \mathbf{A}^T \mathbf{C}^{-1} \mathbf{d}$$

If the matrix on the LHS is invertible, there is a unique solution to the problem

Uniqueness of the solution

Consider the associated eigenvalue problem:

$$(\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B}) \mathbf{m} = \mu \mathbf{m}.$$

If the eigenvalues are all non-zero, then the matrix is invertible

Take inner product with a normalised eigenvector:

$$(\mathbf{A} \mathbf{m})^T \mathbf{C}^{-1} (\mathbf{A} \mathbf{m}) + \lambda \mathbf{m}^T \mathbf{B} \mathbf{m} = \mu,$$

The LHS is positive so long as the trade-off parameter is positive

Regularised least squares is a means of calculating solutions to the inverse problem that also have desirable physical properties. But there are a **lot of arbitrary choices**

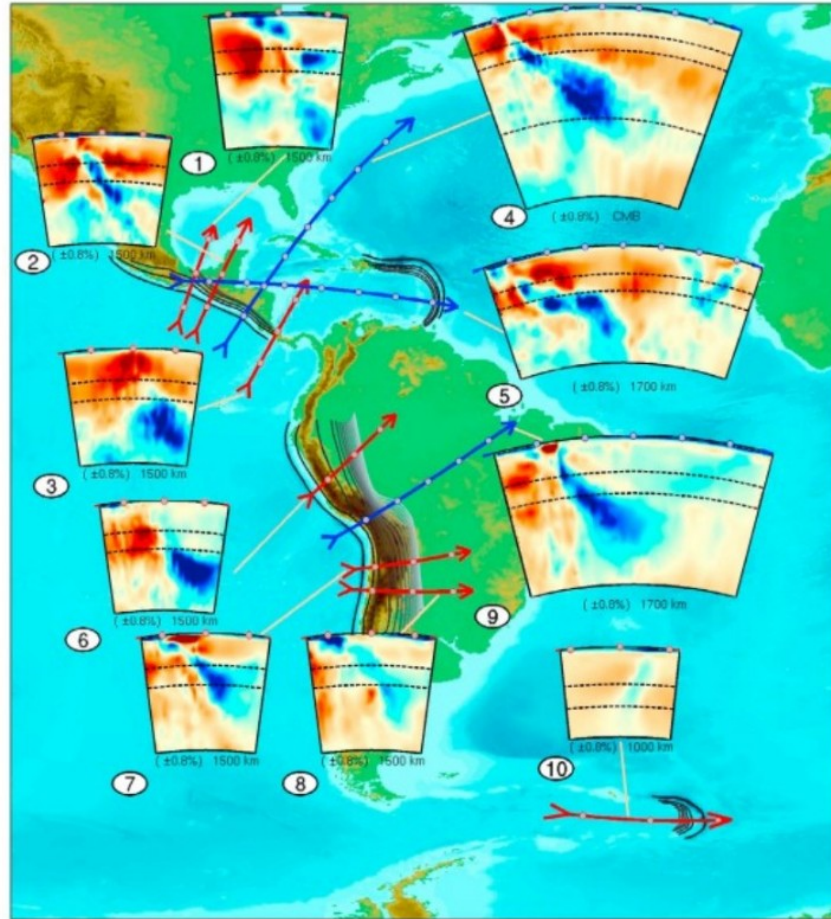
Generalised regularisation

We can also ask that the model stays “close” to a preferred point:

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}(\mathbf{m} - \mathbf{m}_0)^T \mathbf{B}(\mathbf{m} - \mathbf{m}_0),$$

For positive trade-off parameter, we obtain the unique solution:

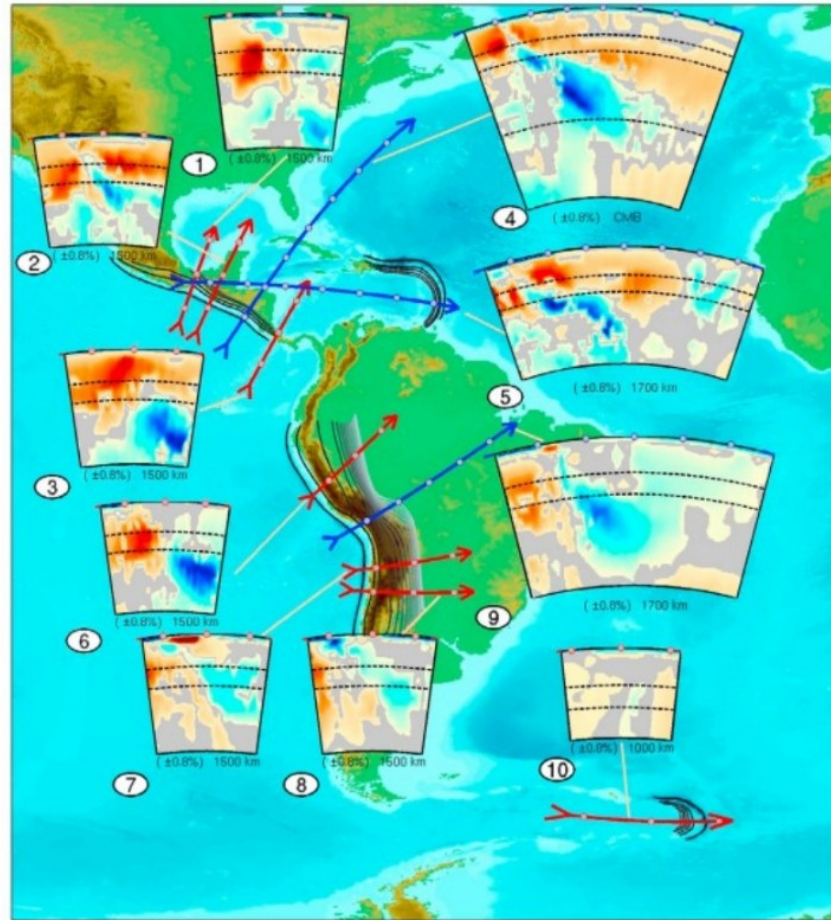
$$\mathbf{m} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{d} + \lambda \mathbf{B} \mathbf{m}_0).$$



De Witt et al. 2012

slow  fast

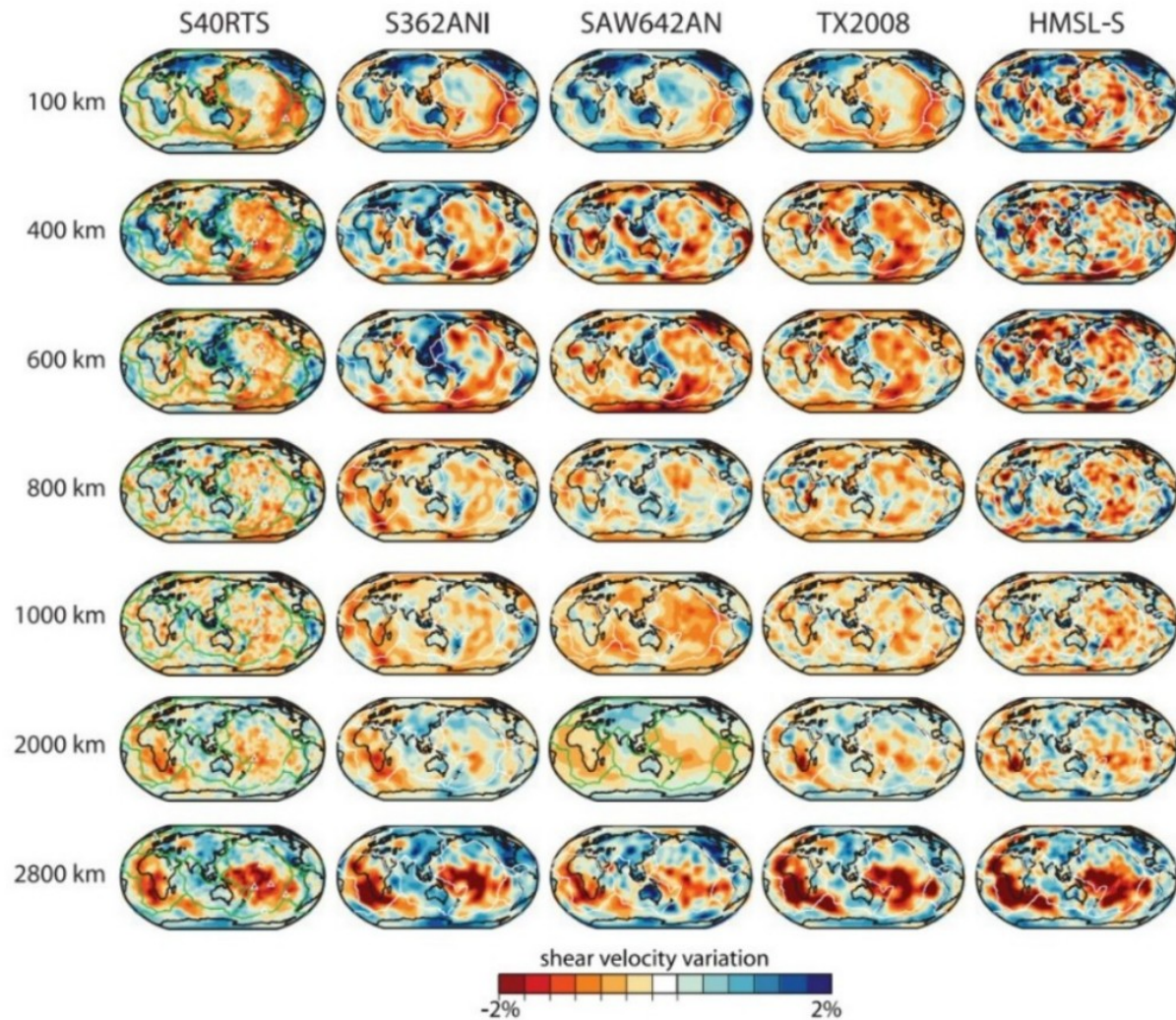
Which features in the model can be believed? Not an easy question!



De Witt et al. 2012

slow  fast

Can vary arbitrary parameters to see what has to be there to fit the data



Different data sets and methods lead to broadly consistent results

A Bayesian perspective

Bayes theorem

Bayes theorem expressed in terms of conditional probability density functions:

$$p(\mathbf{m}|\mathbf{d}) = \frac{p(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{\int p(\mathbf{d}|\mathbf{m})p(\mathbf{m}) d\mathbf{m}'},$$

This is a complete recipe to update prior beliefs given data

But what do probabilities mean in this context?

And how do you arrive at the “correct prior”

Now any decision that depends on the data that is being used in making the inference only requires from the data the posterior distribution. Consequently the problem of inference is effectively solved by stating the posteriori distribution. This is the reason I feel that the basic problem of inference is solved.

D.V. Lindley, *Foundations of Statistical Inference*

I am often asked if the method gives the right answer; or, more particularly, how do you know if you have got the right prior. My reply is that I don't know what is meant by "right" in this context. The Bayesian theory is about coherence, not about right or wrong.

D.V. Lindley. *Bayesian Statistics*

Linear problems with Gaussian priors

Prior and likelihood take the form:

$$p(\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^m \det \mathbf{C}_m}} \exp \left[-\frac{1}{2} (\mathbf{m} - \mathbf{m}_0)^T \mathbf{C}_m^{-1} (\mathbf{m} - \mathbf{m}_0) \right]$$

$$p(\mathbf{d}|\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^n \det \mathbf{C}}} \exp \left[-\frac{1}{2} (\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1} (\mathbf{A}\mathbf{m} - \mathbf{d}) \right].$$

Bayes theorem results in a posterior PDF that is also Gaussian

Posterior mean and covariance

Posterior mean has the same form as a regularised least-squares solution:

$$\mathbf{m}_p = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \mathbf{C}_m^{-1})^{-1} (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{d} + \mathbf{C}_m^{-1} \mathbf{m}_0),$$

Method also provides the posterior covariance:

$$\mathbf{C}_p = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \mathbf{C}_m^{-1})^{-1}.$$

Crucially, this quantifies uncertainty on the model – a combination of propagated data errors and the limitations imposed by the physics of the problem

Bayesian methods have a lot going for them, but they are only as good as the prior